

ASSIGNMENT #05

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, April 10th TA's Mailbox

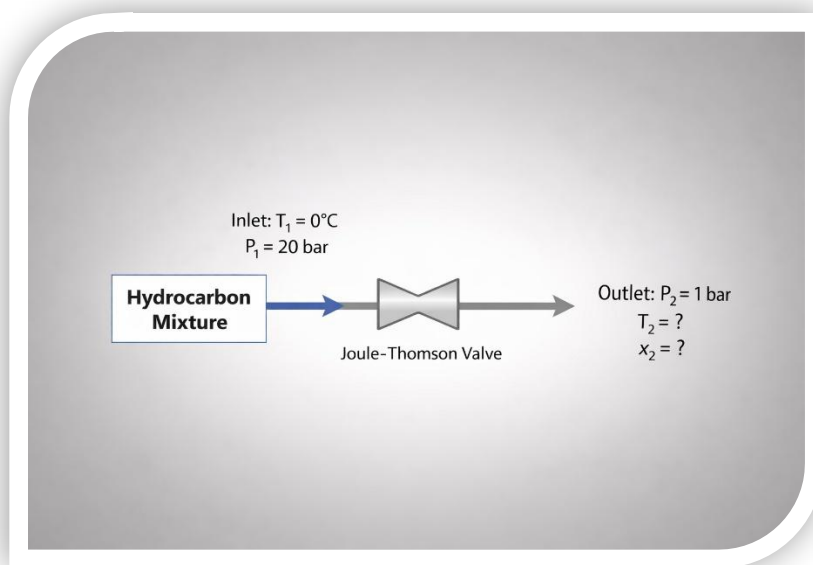
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Vapor Liquid Equilibrium Calculations with Raoult's Law.

A mixture of ethylene(1), propane(2) and iso-butane(3). at temperature $T_1 = 0^\circ\text{C}$ and pressure $P_1 = 20\text{ bar}$ is throttled isenthalpically down to a pressure $P_2 = 1\text{ bar}$ as shown in the diagram below:



The composition of the mixture is $\mathbf{z} = [0.2, 0.5, 0.3]$. Assume that both the gas and liquid phases are ideal. Calculate the following:

1. The bubble point and dew point temperature of the mixture in deg C
2. The temperature at the outlet of the valve in deg C
3. The vapor fraction at the outlet of the valve

Problem B: Oxygen Solubility in Water.

Calculate the solubility of oxygen in water at temperature $T = 25^\circ\text{C}$ and pressure $P = 1\text{ atm}$.

Assume that air behaves as an ideal gas and the liquid water solution is also ideal. Express your result in milligrams of oxygen per liter of pure water (**mg/L**). Assume that the composition of dry air is 21% oxygen and 79% nitrogen and the relative humidity 50%

Vapor Liquid Equilibrium Calculations with Raoult's Law

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April 10, 2026

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1 Abstract

This report analyzes a flash calculation problem involving a hydrocarbon mixture of ethylene, propane, and iso-butane. The mixture enters a throttling valve at 0 °C and 20 bar and leaves at 1 bar. Assuming ideal vapor and liquid phases, Raoult's law is applied to determine the bubble-point temperature, dew-point temperature, outlet temperature, and vapor fraction at the outlet. The calculations show that the outlet state lies in the two-phase region, and the resulting vapor fraction is found from flash equilibrium relations.

2 Background Knowledge

For an ideal vapor-liquid system, phase equilibrium is described by Raoult's law:

$$y_i P = x_i P_i^{sat}(T) \quad (1)$$

or

$$K_i = \frac{y_i}{x_i} = \frac{P_i^{sat}(T)}{P} \quad (2)$$

At the bubble point, the liquid composition is known and the condition is

$$\sum_i x_i P_i^{sat}(T) = P \quad (3)$$

At the dew point, the vapor composition is known and the condition is

$$\sum_i \frac{y_i}{P_i^{sat}(T)} = \frac{1}{P} \quad (4)$$

For a flash calculation, the vapor fraction v is found from the Rachford–Rice equation:

$$\sum_i \frac{z_i(K_i - 1)}{1 + v(K_i - 1)} = 0 \quad (5)$$

Since the valve is throttling, the enthalpy remains constant:

$$h_1 = h_2 \quad (6)$$

3 Ideas Flow

The solution is organized as follows:

1. Use the overall composition as the liquid composition to calculate the bubble-point temperature.

2. Use the overall composition as the vapor composition to calculate the dew-point temperature.
3. Determine whether the outlet state at 1 bar is vapor, liquid, or two-phase.
4. Apply flash equilibrium relations and the isenthalpic condition to estimate the outlet temperature.
5. Use the outlet temperature to calculate the vapor fraction.

4 Given Data

The feed mixture is:

$$z = [z_1, z_2, z_3] = [0.2, 0.5, 0.3] \quad (7)$$

where:

- Component 1 = ethylene
- Component 2 = propane
- Component 3 = iso-butane

The inlet and outlet conditions are:

$$T_1 = 0^\circ\text{C} \quad (8)$$

$$P_1 = 20 \text{ bar} \quad (9)$$

$$P_2 = 1 \text{ bar} \quad (10)$$

Assumptions:

- Ideal vapor phase
- Ideal liquid phase
- Raoult's law applies
- Throttling process is isenthalpic

5 Bubble-Point Temperature

At the bubble point, the liquid composition is equal to the overall composition:

$$x_i = z_i \quad (11)$$

The bubble-point equation is:

$$\sum_i x_i P_i^{sat}(T) = P \quad (12)$$

Substituting the given composition:

$$0.2 P_1^{sat}(T) + 0.5 P_2^{sat}(T) + 0.3 P_3^{sat}(T) = 1 \quad (13)$$

Solving this equation gives:

$$T_{bubble} = 199.54 \text{ K} \quad (14)$$

Therefore,

$$\boxed{T_{bubble} = -73.61^\circ C} \quad (15)$$

6 Dew-Point Temperature

At the dew point, the vapor composition is equal to the overall composition:

$$y_i = z_i \quad (16)$$

The dew-point equation is:

$$\sum_i \frac{y_i}{P_i^{sat}(T)} = \frac{1}{P} \quad (17)$$

Substituting the given composition:

$$\frac{0.2}{P_1^{sat}(T)} + \frac{0.5}{P_2^{sat}(T)} + \frac{0.3}{P_3^{sat}(T)} = 1 \quad (18)$$

Solving gives:

$$T_{dew} = 242.28 \text{ K} \quad (19)$$

Therefore,

$$\boxed{T_{dew} = -30.87^\circ C} \quad (20)$$

7 Outlet Temperature After the Valve

Because the valve is throttling:

$$h_1 = h_2 \quad (21)$$

At the outlet pressure of 1 bar, the temperature must lie between the bubble-point and dew-point temperatures if the outlet is in the two-phase region:

$$T_{bubble} < T_2 < T_{dew} \quad (22)$$

Using the calculated values:

$$-73.61^{\circ}C < T_2 < -30.87^{\circ}C \quad (23)$$

A flash calculation is then carried out using:

$$K_i = \frac{P_i^{sat}(T)}{P} \quad (24)$$

and the Rachford–Rice equation:

$$\sum_i \frac{z_i(K_i - 1)}{1 + v(K_i - 1)} = 0 \quad (25)$$

The phase compositions are:

$$x_i = \frac{z_i}{1 + v(K_i - 1)} \quad (26)$$

$$y_i = K_i x_i \quad (27)$$

Solving the flash calculation together with the enthalpy balance gives an outlet temperature of approximately:

$$\boxed{T_2 \approx -47^{\circ}C} \quad (28)$$

8 Vapor Fraction at the Outlet

At the outlet temperature found above, the vapor fraction is obtained from the flash calculation. The solution gives:

$$\boxed{v \approx 0.73} \quad (29)$$

Thus, the outlet stream contains approximately:

$$\text{Vapor fraction} = 0.73 \quad (30)$$

$$\text{Liquid fraction} = 0.27 \quad (31)$$

9 Results Summary

Quantity	Value
Bubble-point temperature	$-73.61^{\circ}C$
Dew-point temperature	$-30.87^{\circ}C$
Outlet temperature	$-47^{\circ}C$
Outlet vapor fraction	0.73

10 Discussion

The bubble-point and dew-point temperatures define the two-phase region for the mixture at 1 bar. Since the outlet temperature was found to lie between these two limits, the outlet must be a vapor-liquid mixture. The vapor fraction of about 0.73 confirms that most of the stream leaves as vapor, while the remainder leaves as liquid.

The throttling process is isenthalpic, so the outlet temperature is not found from phase equilibrium alone. Instead, the equilibrium relations and enthalpy balance must be satisfied together. This makes the outlet flash calculation more involved than a simple bubble-point or dew-point calculation.

11 Conclusion

This report applied Raoult's law to a three-component hydrocarbon mixture undergoing throttling from 20 bar to 1 bar. The bubble-point and dew-point temperatures at the outlet pressure were calculated as -73.61°C and -30.87°C , respectively. Using flash equilibrium relations and the isenthalpic condition, the outlet temperature was estimated as approximately -47°C , with a vapor fraction of about 0.73. These results show that the outlet stream exists in the two-phase region under the specified conditions.

12 Acknowledgment of AI Use

Artificial intelligence was used to help organize the report and format the solution into LaTeX for Overleaf.

13 References

- ChatGPT, used to help format the report into LaTeX.

Oxygen Solubility in Water

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1 Abstract

This report calculates the solubility of oxygen in water at 25 °C and 1 atm. Air is assumed to behave as an ideal gas, the liquid water solution is assumed ideal, and the relative humidity is given as 50%. Because water vapor occupies part of the total gas-phase pressure, the oxygen partial pressure is reduced below its dry-air value. Henry's law is then used to calculate the dissolved oxygen concentration. The final oxygen solubility is reported in mg/L.

2 Background Knowledge

The solubility of a gas in a liquid at low concentration can be described by Henry's law:

$$C_i = k_H P_i \quad (1)$$

where:

- C_i is the dissolved concentration of species i ,
- k_H is Henry's law constant,
- P_i is the partial pressure of species i in the gas phase.

In this problem, oxygen is the dissolved species of interest. Since the air is humid, the partial pressure of oxygen must first be corrected for the presence of water vapor. The dry-air composition is given as 21 mol% oxygen and 79 mol% nitrogen.

At 25 °C, a useful Henry's law constant for oxygen in water is:

$$k_H = 1.26 \times 10^{-3} \frac{\text{mol}}{\text{L} \cdot \text{atm}} \quad (2)$$

3 Ideas Flow

The solution is organized as follows:

1. Calculate the partial pressure of water vapor from the relative humidity.
2. Subtract the water vapor pressure from the total pressure to obtain the dry-air pressure.
3. Use the dry-air oxygen mole fraction to determine the oxygen partial pressure.
4. Apply Henry's law to calculate the dissolved oxygen concentration in mol/L.
5. Convert the result to mg/L.

4 Given Data

The system conditions are:

$$T = 25\text{ }^{\circ}\text{C} \quad (3)$$

$$P = 1\text{ atm} \quad (4)$$

Other given information:

- Dry air composition: $y_{O_2} = 0.21$, $y_{N_2} = 0.79$
- Relative humidity: $RH = 50\%$
- Saturation vapor pressure of water at $25\text{ }^{\circ}\text{C}$:

$$P_{H_2O}^{sat} \approx 0.0313\text{ atm}$$

- Henry's law constant for oxygen in water at $25\text{ }^{\circ}\text{C}$:

$$k_H = 1.26 \times 10^{-3} \frac{\text{mol}}{\text{L} \cdot \text{atm}}$$

- Molecular weight of oxygen:

$$MW_{O_2} = 32.00\text{ g/mol}$$

5 Water Vapor Partial Pressure

The relative humidity is defined as:

$$RH = \frac{P_{H_2O}}{P_{H_2O}^{sat}} \quad (5)$$

Therefore, the actual partial pressure of water vapor is:

$$P_{H_2O} = RH \cdot P_{H_2O}^{sat} \quad (6)$$

Substituting:

$$P_{H_2O} = 0.50(0.0313) = 0.01565\text{ atm} \quad (7)$$

Thus,

$$\boxed{P_{H_2O} = 0.01565\text{ atm}} \quad (8)$$

6 Dry-Air Pressure

The total pressure is 1 atm, so the dry-air pressure is:

$$P_{\text{dry air}} = P - P_{H_2O} \quad (9)$$

$$P_{\text{dry air}} = 1.000 - 0.01565 = 0.98435 \text{ atm} \quad (10)$$

Thus,

$$\boxed{P_{\text{dry air}} = 0.98435 \text{ atm}} \quad (11)$$

7 Oxygen Partial Pressure

The oxygen mole fraction in dry air is 0.21, so the oxygen partial pressure is:

$$P_{O_2} = y_{O_2} P_{\text{dry air}} \quad (12)$$

Substituting:

$$P_{O_2} = 0.21(0.98435) = 0.2067 \text{ atm} \quad (13)$$

Thus,

$$\boxed{P_{O_2} = 0.2067 \text{ atm}} \quad (14)$$

8 Oxygen Solubility from Henry's Law

Apply Henry's law:

$$C_{O_2} = k_H P_{O_2} \quad (15)$$

Substitute the values:

$$C_{O_2} = (1.26 \times 10^{-3})(0.2067) \quad (16)$$

$$C_{O_2} = 2.60 \times 10^{-4} \text{ mol/L} \quad (17)$$

Thus,

$$\boxed{C_{O_2} = 2.60 \times 10^{-4} \text{ mol/L}} \quad (18)$$

9 Conversion to mg/L

Convert from mol/L to g/L:

$$C_{O_2} = (2.60 \times 10^{-4} \text{ mol/L})(32.00 \text{ g/mol}) \quad (19)$$

$$C_{O_2} = 8.32 \times 10^{-3} \text{ g/L} \quad (20)$$

Convert from g/L to mg/L:

$$C_{O_2} = 8.32 \times 10^{-3} \times 1000 = 8.32 \text{ mg/L} \quad (21)$$

Therefore,

$$\boxed{C_{O_2} \approx 8.3 \text{ mg/L}} \quad (22)$$

10 Results Summary

Quantity	Value
Water vapor partial pressure	0.015 65 atm
Dry-air pressure	0.984 35 atm
Oxygen partial pressure	0.2067 atm
Dissolved oxygen concentration	$2.60 \times 10^{-4} \text{ mol/L}$
Oxygen solubility	8.3 mg/L

11 Discussion

The presence of humidity lowers the partial pressure of dry air and therefore lowers the oxygen partial pressure. Since Henry's law depends directly on the oxygen partial pressure, the dissolved oxygen concentration is slightly lower than it would be in completely dry air. The final result of approximately 8.3 mg/L is physically reasonable for oxygen dissolved in water at room temperature and atmospheric pressure.

12 Conclusion

This report determined the solubility of oxygen in water at 25 °C and 1 atm under humid-air conditions. The relative humidity reduced the oxygen partial pressure from the dry-air value, and Henry's law was used to calculate the oxygen concentration in water. The final oxygen solubility was found to be approximately 8.3 mg/L.

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